Bookmark 1 : High Prior Visits | Bookmark 2 : Extended Stay Risk | Bookmark 3 : Discharge Risk | Bookmark 4 : Med Change Impact | Bookmark 5 : Reset Filters

# Readmission Analysis – All Diagnosis Groups

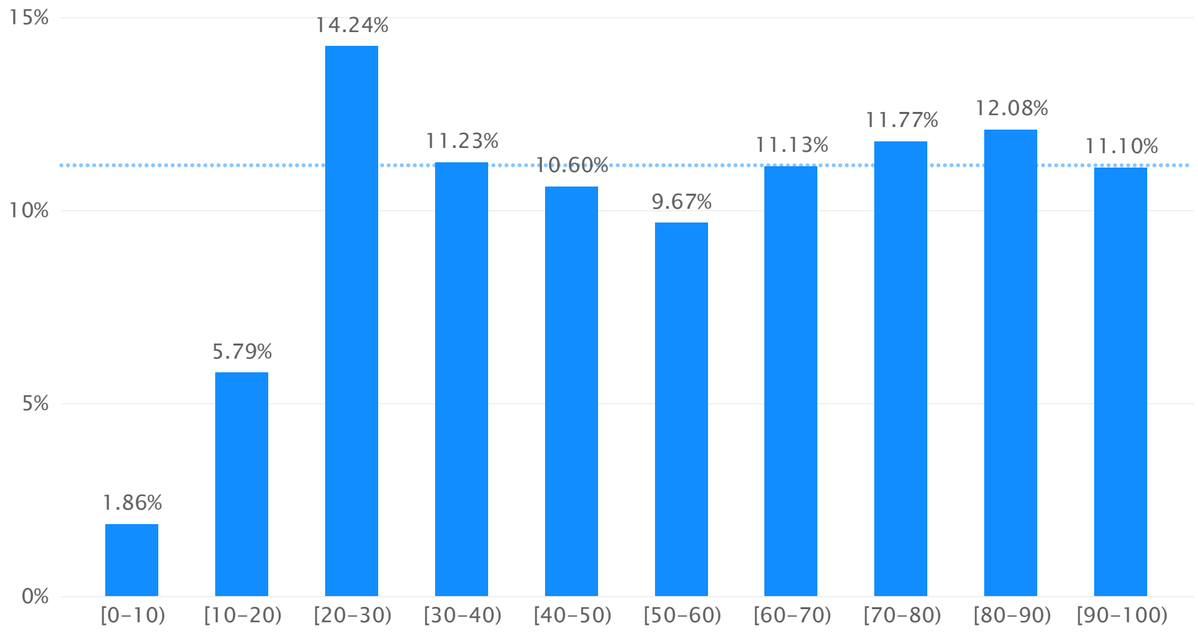
Primary Diagnosis Group: All

Medication Change: All

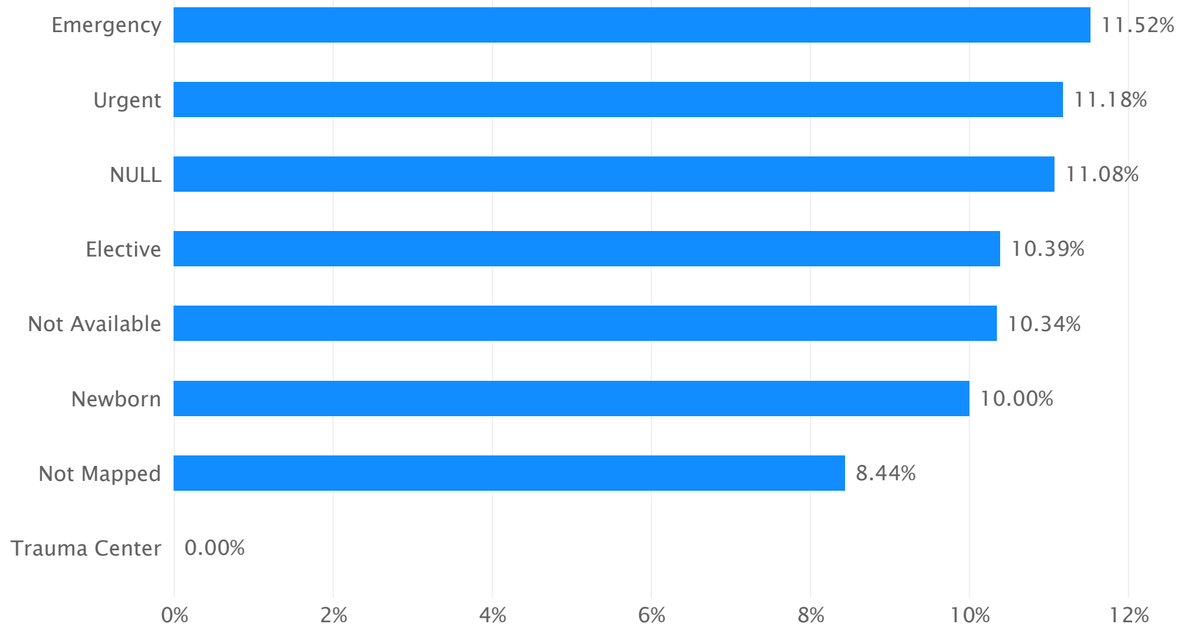
Length of Stay Band: All

| Total Encounters | Readmission Rate (30D) % | Avg Length of Stay (Days) | Avg Medications |
|---|---|---|---|
| 102K | 11.16% | 4.40 | 16.02 |

Readmission Rate (30D) %
by Age Group




Readmission Rate (30D) %
by Admission



Bookmark 1 : High Prior Visits | Bookmark 2 : Extended Stay Risk | Bookmark 3 : Discharge Risk | Bookmark 4 : Med Change Impact | Bookmark 5 : Reset Filters

# Readmission Analysis – All Diagnosis Groups

Primary Diagnosis Group: All

Medication Change: All

Length of Stay Band: All

**Readmission Rate (Med Changed) %**

11.82%

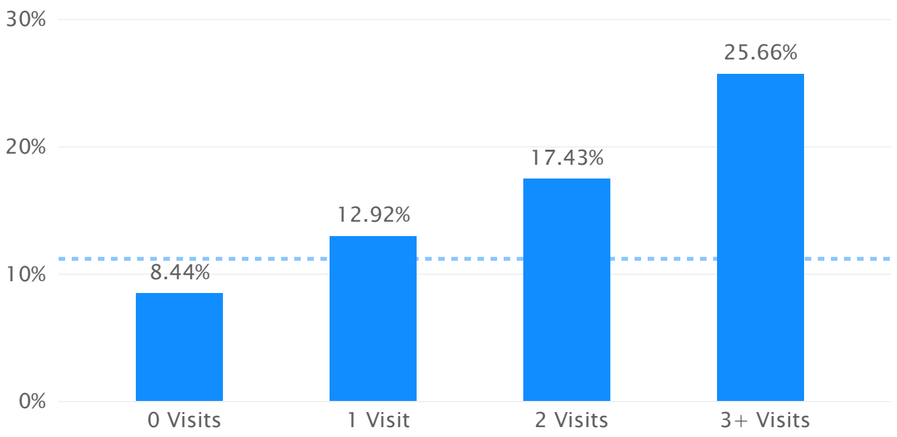
**Readmission Rate (30D) %**

11.16%

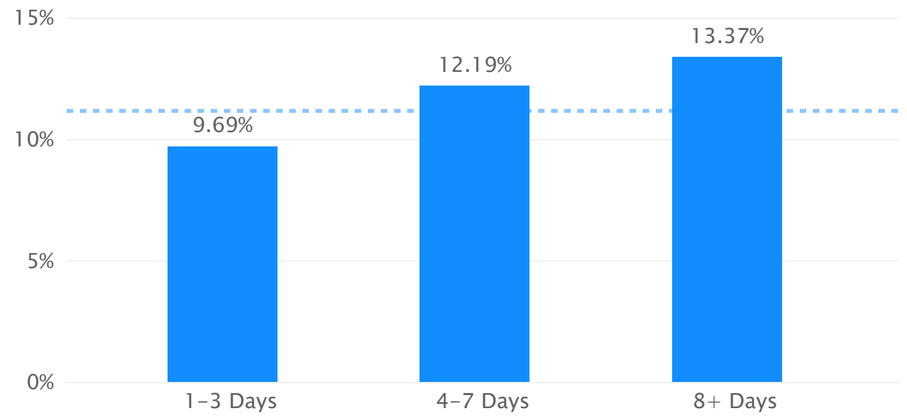
**High Risk Encounters**

1.43K

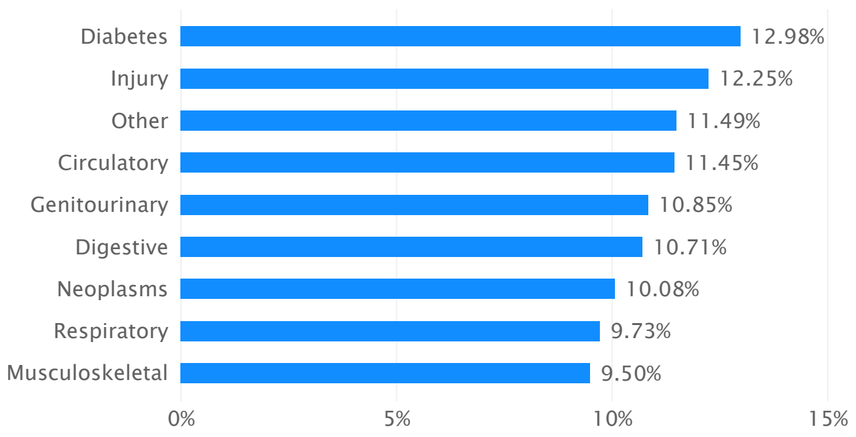
**Readmission Rate (30D) %**
**by Prior Inpatient Visits**



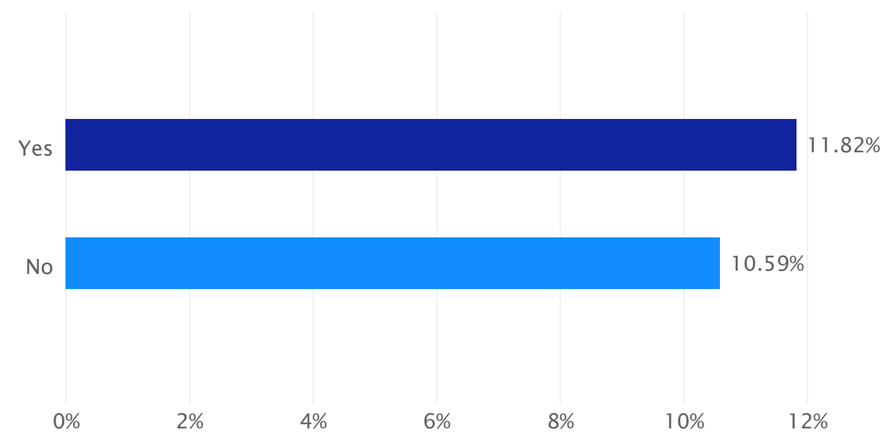

**Readmission Rate (30D) %**
**by Length of Stay Band**




**Readmission Rate (30D) %**
**by Primary Diagnosis Group**




**Readmission Rate (30D) %**
**by Medication Change**



Bookmark 1 : High Prior Visits | Bookmark 2 : Extended Stay Risk | Bookmark 3 : Discharge Risk | **Bookmark 4 : Med Change Impact** | Bookmark 5 : Reset Filters

# Readmission Analysis – All Diagnosis Groups

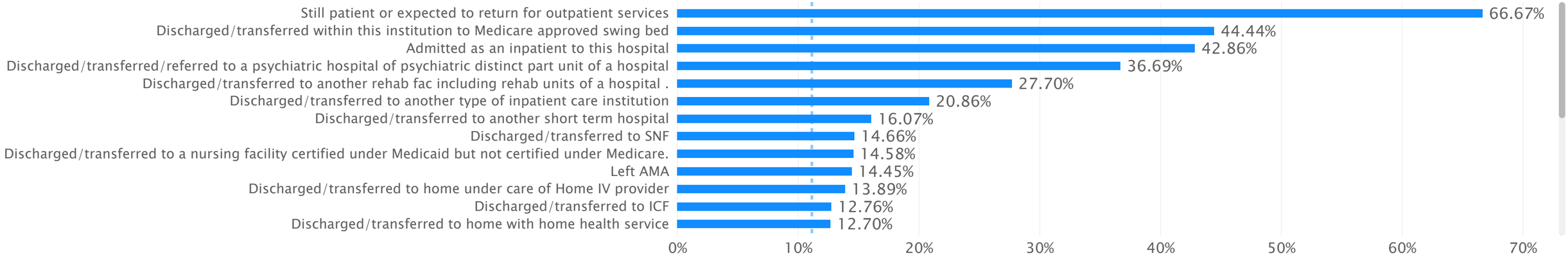
Primary Diagnosis Group: All

Medication Change: All

Length of Stay Band: All

| Discharge Disposition | Admitted as an inpatient to this hospital | | Discharged to home | | Discharged/transferred to a federal health care facility. | | Discharged/transferred to a long term care hospital. | | Discharged/t |
|---|---|---|---|---|---|---|---|---|---|
| Age Group | Readmission Rate (30D) % | Total Encounters | Readmission Rate (30D) % | Total Encounters | Readmission Rate (30D) % | Total Encounters | Readmission Rate (30D) % | Total Encounters | Readmission |
| [0-10) | | | 2.07% | 145 | | | | | |
| [10-20) | 0.00% | 1 | 5.76% | 625 | | | | | |
| [20-30) | | | 13.81% | 1405 | | | 0.00% | 3 | |
| [30-40) | | | 10.42% | 3081 | | | | | |
| [40-50) | 50.00% | 2 | 9.45% | 7448 | | | 3.70% | 27 | |
| [50-60) | | | 7.96% | 12252 | 0.00% | 3 | 10.20% | 49 | |
| [60-70) | 45.45% | 11 | 9.39% | 14437 | 0.00% | 1 | 8.16% | 98 | |
| [70-80) | 33.33% | 3 | 9.39% | 13689 | | | 8.26% | 109 | |
| [80-90) | 50.00% | 4 | 9.90% | 6444 | 0.00% | 1 | 4.90% | 102 | |
| **Total** | **42.86%** | **21** | **9.30%** | **60234** | **0.00%** | **5** | **7.28%** | **412** | |

Readmission Rate (30D) %
by Discharge Disposition



# Readmission Analysis – All Diagnosis Groups

Primary Diagnosis G... All | Medication Change All | Length of Stay Band All

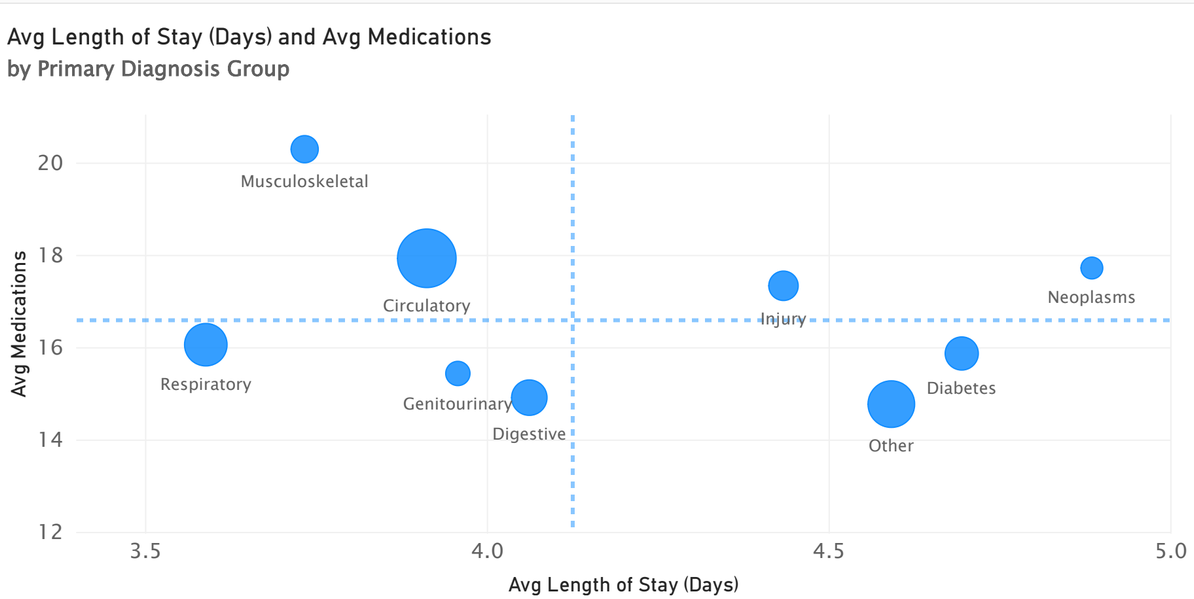
| Age Group | Diagnosis Group | Time in Hospital (Days) | Medications | Lab Procedures | Medication Change | Readmitted Status |
|---|---|---|---|---|---|---|
| [50-60) | Circulatory | 1 | 1 | 30 | No | NO |
| [50-60) | Circulatory | 1 | 1 | 34 | No | NO |
| [50-60) | Circulatory | 1 | 1 | 39 | Yes | NO |
| [50-60) | Circulatory | 1 | 1 | 42 | No | >30 |
| [50-60) | Circulatory | 1 | 1 | 44 | No | NO |
| [50-60) | Circulatory | 1 | 1 | 47 | No | NO |
| [50-60) | Circulatory | 1 | 1 | 53 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 4 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 10 | Yes | NO |
| [50-60) | Circulatory | 1 | 2 | 21 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 31 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 32 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 35 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 40 | No | >30 |
| [50-60) | Circulatory | 1 | 2 | 41 | No | NO |
| [50-60) | Circulatory | 1 | 2 | 42 | No | >30 |
| [50-60) | Circulatory | 1 | 2 | 43 | No | >30 |
| [50-60) | Circulatory | 1 | 3 | 1 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 2 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 3 | No | >30 |
| [50-60) | Circulatory | 1 | 3 | 8 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 21 | No | <30 |
| [50-60) | Circulatory | 1 | 3 | 33 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 34 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 40 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 41 | No | NO |
| [50-60) | Circulatory | 1 | 3 | 44 | Yes | >30 |
| **Total** | | | | | | |

**High-Risk Segment Share %**
1.38%

**Avg Length of Stay (Days)**
4.13

**High Risk Encounters**
238

**Readmission Rate (30D) %**
9.67%



# Executive Summary & Strategic Clinical Insights

## Core Findings

### High Inpatient Utilization (Primary Risk Driver)

- **Metric:** 3+ Prior Inpatient Visits -> **25.66% Readmission Rate**
- **Takeaway:** Frequent acute care utilization is the single strongest predictor of 30-day return. Patients with 3+ prior visits readmit at more than double the 11.16% baseline.

### High-Risk Post-Acute Discharge Pathways

- **Metric:** Transfers to Outpatient/Inpatient Facilities -> **42%-66% Readmission Rate**
- **Takeaway:** Transitions of care to post-acute care and outpatient facilities represent severe vulnerability. Discharge coordination failure is a major contributor to early returns.

### Complex Care & Extended Length of Stay

- **Metric:** Hospitalization >8 Days -> **13.37% Readmission Rate**
- **Takeaway:** Patient encounters requiring extended stays (>8 days) or high medication counts (>16 medications) show compounding complexity and higher relapse rates.

### Medication Adjustment Vulnerability

- **Metric:** Regimen Changes -> **11.82% Readmission Rate**
- **Takeaway:** Patients undergoing active medication regimen adjustments during admission carry elevated risk, pointing to post-discharge prescription adherence gaps.

## Operational Action Plan

| Focus Area | Operational Initiative | Targeted Metric | Impact Goal |
|---|---|---|---|
| **Complex Transitions** | Mandatory 48-hour post-discharge telehealth check-in for SNF/Rehab transfers. | Discharge Disposition | Reduce post-acute return rate by 15% |
| **High Utilizers** | Enroll patients with $\geq 3$ prior inpatient visits into dedicated nurse case management. | Prior Inpatient Visits | Lower 30-day return rate to $< 18\%$ |
| **Medication Reconciliation** | Pharmacist-led discharge medication review for all patients with regimen changes. | Medication Change | Mitigate medication-related readmissions |

# Data Governance, Schema Architecture & Methodology

## Data Provenance & Scope

- **Source Dataset**: De-identified Clinical Diabetes Inpatient Dataset (diabetic_data.csv).
- **Observation Period**: 10-year period (1999–2008) across 130 US hospitals.
- **Volume**: 101,766 total patient encounters.
- **Primary Outcome Variable**: 30-Day Readmission Flag (Readmitted Status = <30).

## Data Model Architecture

- **Schema Pattern**: Star Schema centered on Fact_Encounters.
- **Relationships**: 1-to-many single-direction relationships (1 to *) linked via surrogate key integer columns:
  - Dim_AdmissionType[AdmissionTypeKey]
  - Dim_DischargeDisposition[DischargeDispositionKey]
  - Dim_AdmissionSource[AdmissionSourceKey]
- **DAX Layer**: Isolated _Measures table housing all explicit measures; all implicit visual summation disabled.

## Data Cleaning & ETL Rules (Power Query UI)

- **Missing Value Handling**: Replaced missing/unknown string values (?, NULL) with standard categories ("Unknown", "Not Available").
- **Column Exclusions**: Removed high-null features with >40% missing values (weight, payer_code, medical_specialty).
- **Calculated Feature Bins**: Added custom conditional columns for:
  - Age Group ([0-10) through [90-100))
  - Length of Stay Band (1-3 Days, 4-7 Days, 8+ Days)
  - Prior Inpatient Visits Band (0 Visits, 1-2 Visits, 3+ Visits)

## Patient Privacy & Data Compliance

- **De-identification**: All Direct Identifiers (Patient Names, SSNs, exact admission dates) were stripped prior to model ingestion.
- **Surrogate Identification**: Encounters and individual patients are referenced exclusively via de-identified IDs (Patient ID, Encounter ID).
- **HIPAA Alignment**: Dataset complies with Safe Harbor de-identification standards under HIPAA regulations.